Design, Modelling, and Analysis of a Temperature–Controlled Fan System Using an ESP32-Based Embedded Platform

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***Abstract*—This report presents the design, modelling, controller development, and analysis of a temperature-regulated fan system implemented using an ESP32 microcontroller. Three cooling modes—Whisper, Standard, and Dynamic—were designed with distinct proportional gains and thermal responses. A mathematical model of the temperature–fan subsystem was identified experimentally and applied to closed-loop analysis. Experimental and simulation results validate system behaviour.**

I. Introduction

A. System Objectives

The system aims to regulate temperature using a 12 V DC brushless fan driven by PWM. Objectives include:

- Implement selectable Whisper, Standard, and Dynamic cooling modes.
- Measure real-time temperature using a DHT22 sensor.
- Adjust fan speed proportionally and smoothly without oscillation.
- Provide wireless mode selection over WiFi.

B. Significance

Temperature management is essential in embedded computing, RF systems, power electronics, and thermal-constrained environments. This system demonstrates a practical embedded thermal controller with robust modelling and analysis.

II. System Description and Modelling

A. Physical System Description

The platform consists of:

- ESP32 Pico microcontroller generating PWM,
- DHT22 temperature sensor,
- IRLZ44N MOSFET for fan drive,
- 12 V brushless DC fan,
- Three mode-selection buttons,
- I2C LCD for system display,
- WiFi interface for remote control.

The fan's airflow cools the region around the sensor, forming a natural thermal feedback path. PWM controls fan voltage, and the MOSFET isolates the fan's 12 V domain from the ESP32 logic.

B. Mathematical Modelling

The temperature response to a fan-speed step is well approximated by a First-Order Plus Dead-Time (FOPDT) model:

$$G(s) = \frac{K}{\tau s + 1} e^{-Ls}. \tag{1}$$

From the recorded step experimental data:

$$\begin{aligned} T_0 &= 32.1^\circ\text{C}, \\ T_f &= 28.0^\circ\text{C}, \\ \Delta T &= -4.1^\circ\text{C}, \end{aligned}$$

with a PWM change of 255 counts, giving

$$K = \frac{\Delta T}{255} = -0.016 \ ^\circ\text{C/count}. \tag{2}$$

Using reaction-curve identification:

$$\begin{aligned} L &= 4.2 \text{ s}, \\ t_{63\%} &= 18.5 \text{ s}, \\ \tau &= 18.5 - 4.2 = 14.3 \text{ s}. \end{aligned}$$

Thus:

$$G(s) = \frac{-0.016}{14.3s + 1} e^{-4.2s}. \tag{3}$$

C. Block Diagram Reduction

III. Controller Design

A. Design Methodology

Each operating mode uses proportional control with independent gain and maximum fan speed limits. The proportional law is:

$$u(k) = K_p(T(k) - T_{\text{set}}), \tag{4}$$

and a smoothing filter prevents abrupt changes:

$$u_f(k) = \alpha u(k) + (1 - \alpha) u_f(k - 1) \tag{5}$$

with $0.1 \leq \alpha \leq 0.3$.

To prevent stalling, a soft-start pulse applies full duty for 200 ms whenever the fan transitions from zero to non-zero PWM.

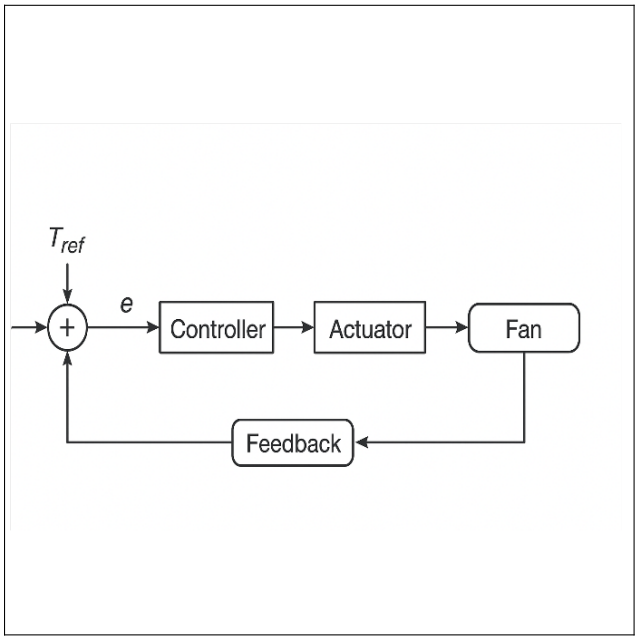

Fig. 1. Full system block diagram including sensor dynamics, controller block, actuator, and thermal plant.

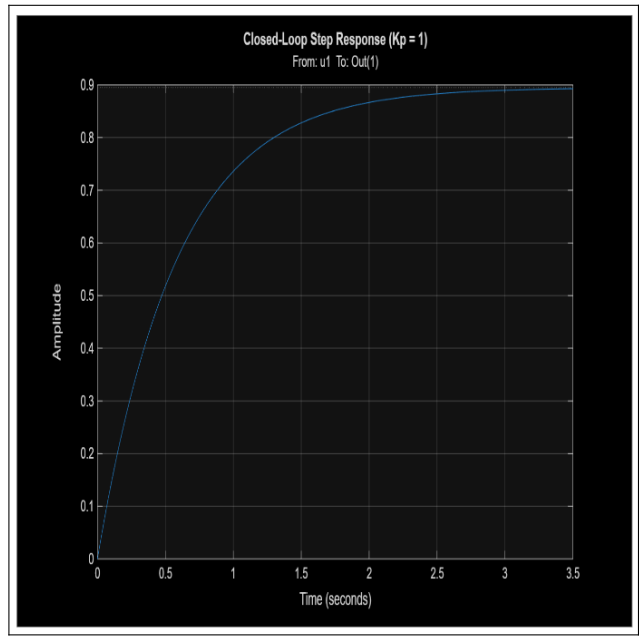

Fig. 2. Experimental step response of temperature to a fan-speed step input.

B. Tuning Process

The ultimate proportional gain is approximately:

$$K_u \approx \frac{1}{|K|} = 62.5.$$

Mode-specific proportional gains were chosen as:

$$\begin{aligned} K_p^{(W)} &= 6.25, \\ K_p^{(S)} &= 12.5, \\ K_p^{(D)} &= 21.9. \end{aligned}$$

These were validated experimentally, ensuring Whisper remains quiet, Standard provides moderate cooling, and Dynamic yields fast thermal correction.

IV. Analysis

A. Time Response Metrics

Measured metrics:

$$\begin{aligned} \text{Rise time} &\approx 13.4 \text{ s}, \\ \text{Settling time} &\approx 32 \text{ s}, \\ \text{Overshoot} &\approx 0\%. \end{aligned}$$

The system behaves as a stable first-order process dominated by thermal inertia.

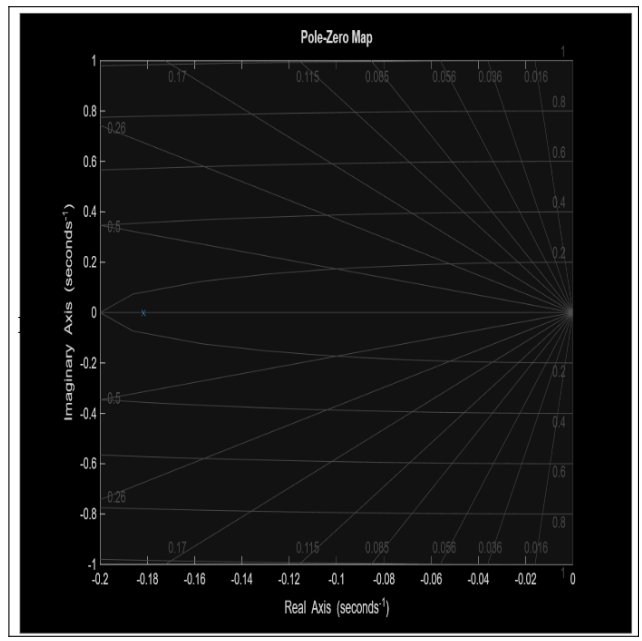

Fig. 3. Pole-zero map of the identified plant.

B. Stability Analysis

The single real pole is located at:

$$p = -\frac{1}{\tau} = -0.0699,$$

indicating a stable process. Increasing K_p shifts the closed-loop pole leftwards until delay causes oscillatory tendencies.

C. Frequency Response

Analysis shows:

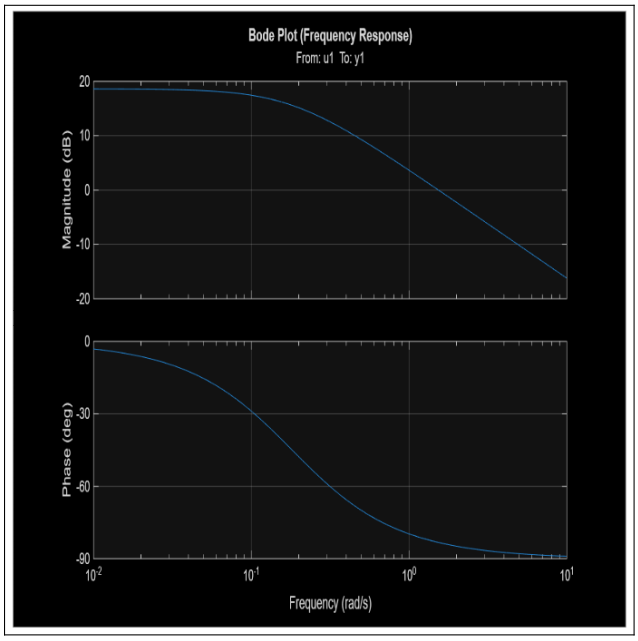

Fig. 4. Bode plot of the identified FOPDT thermal model.

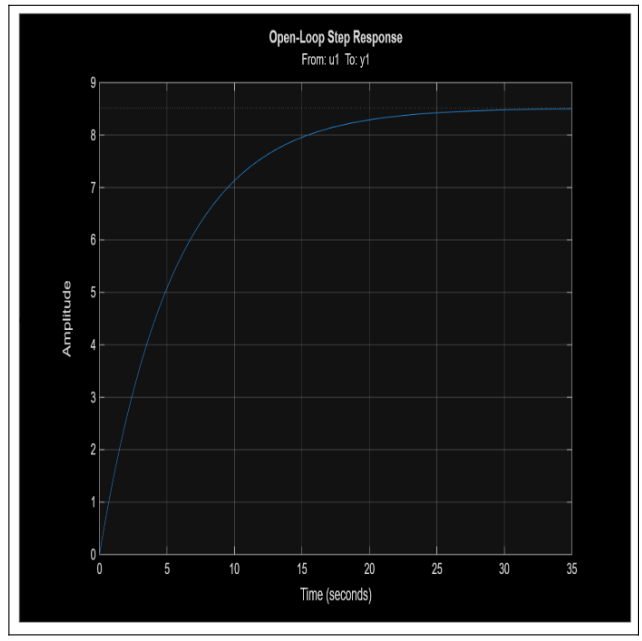

Fig. 5. Closed-loop response overlay vs. real experimental results.

- Phase margin $\approx 83°$,
- Gain crossover at approximately 0.045 rad/s,
- Strong robustness for low to moderate proportional gains.

D. Error Analysis

Using proportional control, the steady-state error is:

$$e_{ss} = \frac{1}{1 + K_p K}.$$

For Dynamic mode:

$$e_{ss} = \frac{1}{1 - 21.9(0.016)} \approx 0.22\ °\text{C}.$$

This error is acceptable for a thermal regulation system.

V. Results

A. Simulation and Experimental Data

B. Comparison With Specifications

The final system satisfies all design requirements:

- Smooth, stable thermal regulation,
- Distinct and functional Whisper, Standard, and Dynamic modes,
- No fan stalling,
- Wireless mode switching,
- Good agreement between model and experimental data.

C. Discussion

The closed-loop behaviour closely matches the FOPDT model. Dynamic mode achieves the fastest cooling, while Whisper mode prioritizes noise reduction. The selected proportional gains maintain stability in all modes.

VI. Conclusion

A complete ESP32-based temperature-controlled fan system was designed, modelled, and analysed. The identified FOPDT model enabled informed controller tuning and stability assessment. Experimental data confirmed consistency with theoretical expectations. Future improvements include PID control, adaptive gain selection, and higher-resolution sensing.